



Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

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Parallel Computing– II part–Thursday, June 30th, 2023

Polimi ID _____

Surname _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:15m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 3 (4 points) _____

Exercise n. 1

Answer the following questions about parallel patterns and briefly explain (without an explanation, the answer will be considered invalid)

A. Please define what a parallel pattern scan is: assumptions, complexity, and example. (1)

B. Please describe how permutation scatter works? (1)

C. Please describe how the Expand pattern works. (1)

D. Can you remove a read after write dependency? (1)

Exercise n. 2

Answer the following questions about parallel algorithms and OpenMP (without an explanation, the answer will be considered invalid).

- A. Describe the method to qualitatively evaluate a parallel algorithm through its data flow graph. (2)

- B. Describe how the scope of variables can be specified upon entering an OpenMP parallel region. (2)

Exercise n. 3

Answer the following questions about GPGPU computing (without an explanation, the answer will be considered invalid).

- A. What are the characteristics of a program that make it amenable to parallelization on a GPU? (1)

- B. What are the advantages of using asynchronous memory prefetching when transferring data from the host to a device? (1)

- C. Is an execution configuration of $\langle\langle\langle 100, 1000 \rangle\rangle\rangle$ suitable for a device with 40 Streaming Multiprocessors and 32-thread warps? (1)

- D. Describe the types of memory available on NVIDIA GPUs and who can access each of them. (1)

Exercise n. 4

Answer the following questions about Halide (without an explanation, the answer will be considered invalid).

- A. Describe the trade-offs that can be explored in the design space created by the Halide scheduling directives. (2)

- B. Describe what is the effect of the following Halide schedule: (2)

```
gradient.split(x, x_outer, x_inner, 4);  
gradient.split(y, y_outer, y_inner, 4);  
gradient.reorder(x_inner, y_inner, x_outer, y_outer);
```
